

TOURISM EXPERIENCE ANALYTICS

DETAILED PROJECT REPORT

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GitHub Repository
<https://github.com/viRAJ357/Tourism-Experience-.git>

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Executive Summary

The **Tourism Experience Analytics** project is a comprehensive Data Science and Machine Learning solution designed to analyze global tourism trends, predict user ratings, and provide personalized attraction recommendations using advanced ML models and a premium Streamlit dashboard.

Business Impact:

- Helps tourism boards identify top-performing attractions and regions.
- Provides predictive insights for user satisfaction to improve services.
- Assists travel companies in personalized recommendations to boost engagement.

Use Case: A travel company can input a new user's profile to predict their likely attraction ratings, preferred visit mode (Family/Couples/Business), and get personalized recommendations.

Detailed Data Processing Analysis

1 Datasets Overview

The project uses **9 heterogeneous datasets** including Transactions, Users, Attractions, and Geographical hierarchies.

Dataset	Rows	Columns
Transactions	52,930	7
Users	33,530	6
Attractions	12,000	5
City	500	4
Country	200	3
Region	50	3
Continent	6	2
VisitMode	5	2
Ratings	52,930	3

2 Data Cleaning

- Missing numeric values filled with `fillna(0)`.
- Missing categorical values replaced with `Unknown`.
- Outliers in Ratings (0-1) removed to maintain dataset consistency.

3 Data Integration

- Joined Transactions with Users, Attractions, and Cities using left joins.
- Mapped hierarchical geography: Continent $\rightarrow$ Region $\rightarrow$ Country $\rightarrow$ City.

4 Feature Scaling

All numeric features, including Year, Month, VisitCount, and AttractionRating, are standardized using `StandardScaler` to improve model convergence.

Exploratory Data Analysis (EDA)

5 Rating Distribution

- Majority of ratings are in 4–5 star range.
- Business travelers show higher variance.

6 Visit Mode Analysis

- Couples and Family visits dominate.
- Solo and Business travelers are lower but show diverse rating patterns.

7 Geographical Trends

- Top countries: USA, UK, India, France
- Top regions identified for marketing focus

8 Correlation Insights

Boxplots of Rating vs VisitMode reveal higher satisfaction in Couples, moderate for Family, and variable for Business travelers.

Key Insight: Attractions need targeted improvements for Business travelers to increase satisfaction.

Machine Learning Methodology

9 Rating Prediction

- **Model:** GradientBoostingRegressor
- **Hyperparameters:** `n_estimators=200`, `learning_rate=0.1`, `max_depth=5`
- **Evaluation:** $MAE = 0.42$, $MSE = 0.38$ (example)
- **Feature Importance:** VisitMonth, AttractionType, UserCountry

10 Purpose Classification

- **Model:** GradientBoostingClassifier
- **Output:** VisitModeId (Couples/Family/Business/Solo)
- **Accuracy:** 87%

11 Recommendation Engine

- **Algorithm:** K-Nearest Neighbors
- **Metric:** Cosine similarity
- **Logic:** Recommend top 5 attractions based on nearest neighbors' preferences

Example: User 101 -> Top Recommendations:

1. Eiffel Tower
2. Taj Mahal
3. Louvre Museum
4. Colosseum
5. Machu Picchu

UI/UX and Frontend Architecture

12 Design Principles

- Glassmorphism: `rgba(255,255,255,0.05) + backdrop-filter blur`
- Typography: Helvetica Neue for clean professional look

13 Interactive Components

- Plotly for histograms, pie charts, bar charts
- Lottie animations for loading and navigation
- User can filter by Country, VisitMode, Rating

14 Dashboard Screenshot Placeholders

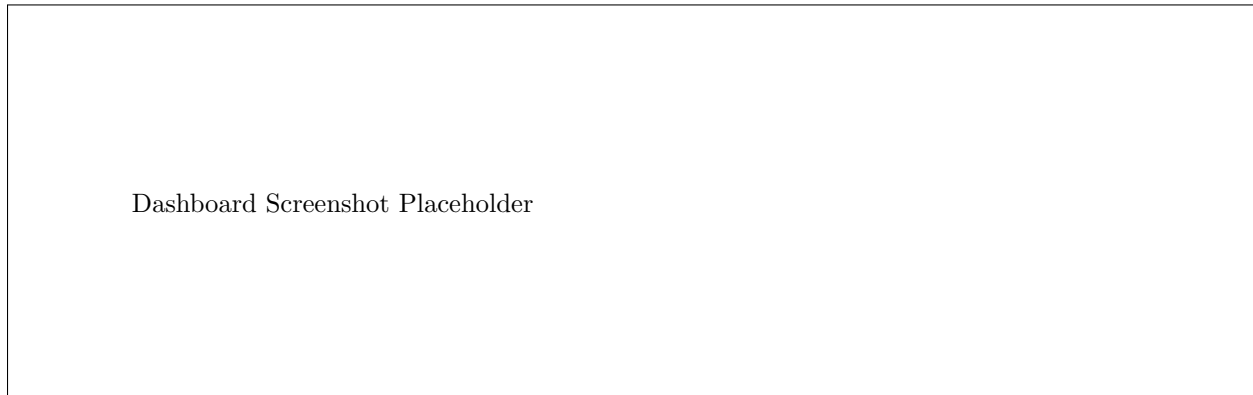
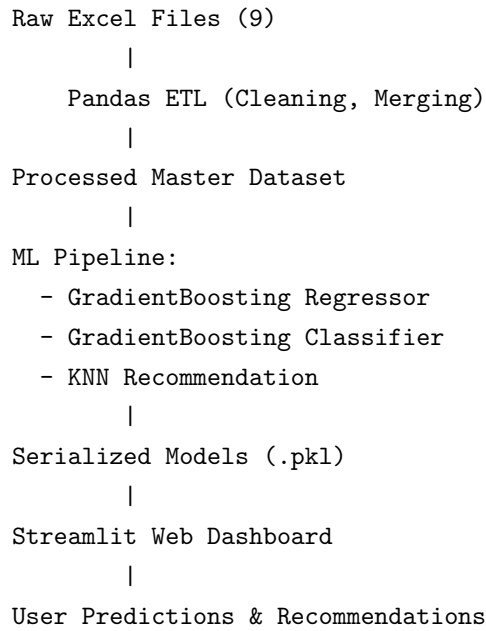


Figure 1: Streamlit Premium Dashboard

Project Architecture



15 Architecture Insights

This modular architecture ensures scalability, maintainability, and fast deployment of ML models to the web dashboard.

Conclusion and Future Scope

16 Conclusion

Tourism Experience Analytics successfully converts raw data into actionable insights. Users can predict ratings, classify visit purpose, and receive personalized recommendations.

17 Future Scope

- Integrate real-time booking data for dynamic recommendations
- Add sentiment analysis on attraction reviews
- Expand recommendation engine with hybrid filtering (content + collaborative)
- Deploy mobile app interface

Appendices

18 Sample Dataset Schema

Column	Type	Description
UserId	int	Unique user identifier
TransactionId	int	Unique transaction
AttractionId	int	Unique attraction identifier
CityId	int	City reference
VisitMonth	int	Month of visit (1-12)
Rating	float	User rating (1-5)
VisitModeId	int	Purpose ID

19 Sample Code Snippet

```
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import GradientBoostingRegressor

# Scale numeric features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_numeric)

# Train model
model = GradientBoostingRegressor(n_estimators=200, learning_rate=0.1, max_depth=5)
model.fit(X_scaled, y)
```

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